

Python Final Review – Summary

1. Basic Python Program: Average Speed Example

To create a simple program that calculates average speed, the computer must be instructed to:

- Accept a number representing the distance.
- Accept a number representing the time.
- Divide distance by time and store the result.
- Display the result in a readable format.

Computers can perform calculations, but they need precise instructions to understand what the numbers represent.

Natural Languages vs. Programming Languages – Summary

A language is a tool for expressing thoughts. Humans use many forms of language, such as body language and spoken languages, to communicate and think. Computers also use languages, but theirs are artificial—called **machine languages**—made up of simple, predefined commands forming a **instruction list (IL)**.

Unlike humans, computers have no intelligence and cannot create new languages. Human (natural) languages, on the other hand, developed and evolved naturally, constantly changing as new words appear and old ones fade.

2. Understanding Programming Languages

A programming language is a tool for expressing instructions to the computer. Like human languages, programming languages include:

- **Alphabet:** Symbols used to form code.
- **Lexis (Vocabulary):** Valid keywords and names.
- **Syntax:** Rules that define the structure of code.
- **Semantics:** Meaning behind the code instructions.

Computers use machine language (binary), which is difficult for humans to write and understand. High-level languages like Python serve as a bridge between human logic and machine operations.

3. Source Code and Compilation vs Interpretation

Source Code

- Code written by the programmer in a high-level language.
- Stored in a plain text file, called a source file.

Two Ways to Execute Code

1. Compilation:

- Translates source code into machine code once.
- Produces a file (e.g., .exe) that can be run on a computer without needing the source code or compiler again.
- Examples: C, C++

2. Interpretation:

- Translates and runs code **line by line** every time it is executed.
- Requires the interpreter to be installed on the user's machine.
- Examples: Python, JavaScript

4. Advantages and Disadvantages

Compilation

Advantages:

- Faster execution after compilation.
- End-users do not need the compiler.
- Machine code is difficult to reverse-engineer, keeping your logic secure.

Disadvantages:

- Compilation takes time.

- A separate compiler is needed for each platform.

Interpretation

Advantages:

- No need to compile before running.
- Portable across platforms, as long as the interpreter is available.

Disadvantages:

- Slower performance during execution.
 - Both developer and user must have the interpreter installed.
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5. Python Language Overview

- Python is an interpreted, object-oriented, high-level programming language with dynamic semantics.
 - Created by Guido van Rossum in the Netherlands.
 - The name "Python" comes from the British comedy group *Monty Python*, not the snake.
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6. Goals of Python (1999, Guido van Rossum)

- Simple and intuitive, yet powerful.
- Open-source and community-driven.
- Code should be as readable as English.

- Suitable for everyday tasks with quick development time.
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7. Why Learn Python?

- Easy to learn and teach.
 - Encourages clean and readable code.
 - Supports rapid development.
 - Free, open-source, and cross-platform.
 - Widely used in industry, research, and education.
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8. Where Python is Used

- Web development
 - Automation and scripting
 - Data science and machine learning
 - Cloud services and APIs
 - Testing and software tools
 - Education
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9. Where Python is Not Common

- Low-level system programming (drivers, kernels)

- Mobile app development (limited support, though improving)
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10. Python Versions

Python 2:

- Legacy version, still in use in old systems.
- Not compatible with Python 3.

Python 3:

- Current standard.
- Improved syntax and features.
- Not backward compatible with Python 2.

To run Python 2 code in Python 3, it must be rewritten or carefully migrated.

11. Python Implementations

CPython:

- The **official and most widely used** implementation.
- Written in C.
- Maintained by the Python Software Foundation (PSF).

Jython:

- Python implemented in Java.

- Integrates well with Java-based systems.
- Currently supports only Python 2.

PyPy:

- Python written in a subset of Python called RPython.
- Faster than CPython due to Just-In-Time (JIT) compilation.
- Useful for testing new features and improving performance.

Cython:

- A tool that lets you write Python code, then convert it to C for performance.
- Often used for scientific computing and performance optimization.

12. Tools for Python Development

To start programming in Python, you need:

- **Editor:** For writing code (e.g., VS Code, PyCharm).
- **Console/Terminal:** To run programs and interact with the system.
- **Debugger:** To run code step-by-step and identify bugs.
- **IDLE:** The default IDE included with Python installation. Stands for *Integrated Development and Learning Environment*.
